

Calit Erdem Anil - Math 572

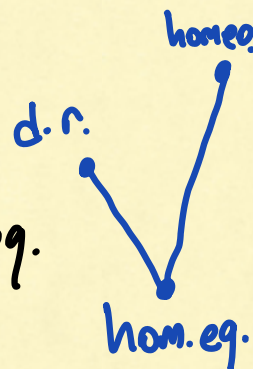
Presentation,

Hatcher (prop 1.18-1.19)

Q: Do similar things induce similar things?

BIG THEM: X, Y homotopy equivalent,
then their fundamental groups are isomorphic,
for some fixed basepoint and h.equivalence f ! i.e.
 $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.

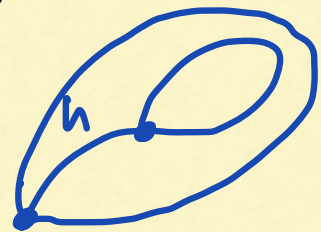
Outline is showing for homeomorphism, def. ret., hom. eq.



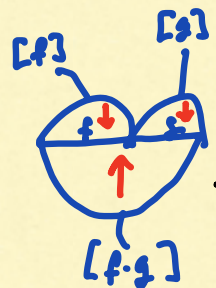
Before that,

Recall: Induced homomorphism, for some path h
connecting x_0, x_1 ,

$$\beta_h: \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$$
$$[f] \mapsto [h f \bar{h}]$$



is an isomorphism since $[f][g] = [fg]$,
i.e. concatenation is well defined for path-homotopic loops,



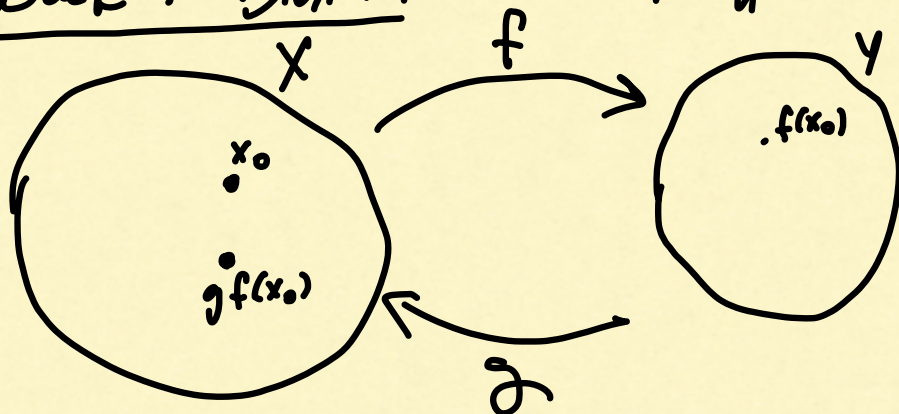
- $(fg)_* = f_* g_*$

- $(fg) = Id_X \Rightarrow (fg)_* = (Id_X)_* = Id_{\pi}$

One should immediately ask, what about $(fg) \circ_h (Id)$?

and this is our question.

Back to BlgThm: let $X \simeq_h Y$. f, g hom. equivalence.



$$(fg) \simeq Id_X, (fg) \neq Id_X$$

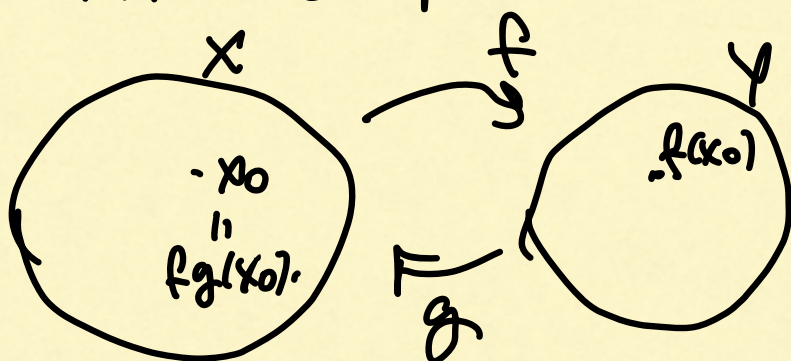
(can we connect basepoints, if so how?)

(Spoiler: homotopy!)

(1) homeo.: X, Y homeomorphic.

d.r.

hom. eq.

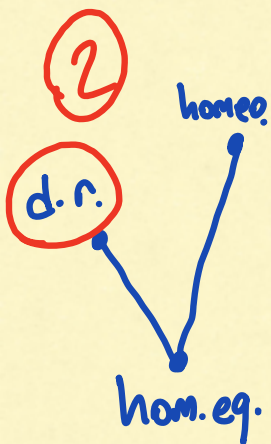


basepoint remains the same, but do they, sf induce some isomorphism btwn fundamental groups $\pi_1(X, x_0), \pi_1(Y, f(x_0))$? Yes since,

$$(gf) = Id_X \Rightarrow (gf)_* = Id_{\pi} \text{ and } g_* f_* = (gf)_*$$

$$\pi_1(X, x_0) \xrightarrow{f_*} \pi_1(Y, f(x_0)) \xrightarrow{g_*} \pi_1(X, x_0), \text{ isomorphism.}$$

Thus, f_* injective. Doing same for $(fg)_*$, f_* surjective.



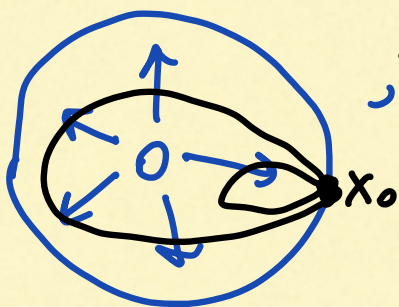
$A \subseteq X$, deformation retraction of X .

i.e. $\exists r: X \rightarrow A, i: A \rightarrow X$

$r(r(x)) = r(x)$, constant on it's image (on A).

and for some homotopy connecting r, i (say r_t)
 r_t fixes A pointwise for each t !

and $ri = Id_A, ir \simeq_h Id_X$



To be strategic, we may choose basepoint
 on X_0 . So far pushing loops of X
 to A is immediate through homotopy
 but for basepoints we need some time.

$$\pi_1(A, x_0) \xrightarrow{i_*} \pi_1(X, x_1) \xrightarrow{\hat{r}_*} \pi_1(A, x_1)$$

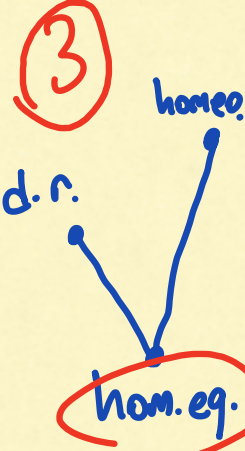
$(ri)_* = Id_{\pi_1}$ so i_* injective. For the surjectivity,

let g be some loop of X at x_1 , $[g] \in \pi_1(X, x_1)$

$r_t([g])$ connects our loop to loops in A at x_0 .

Then $i_*: \pi_1(A, x_0) \rightarrow \pi_1(X, x_0)$ surjective
 $[f] \mapsto [i \circ f]$

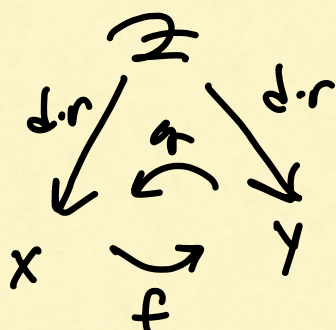
as ANY, loop in X is path-homotopic to loops
 in A !



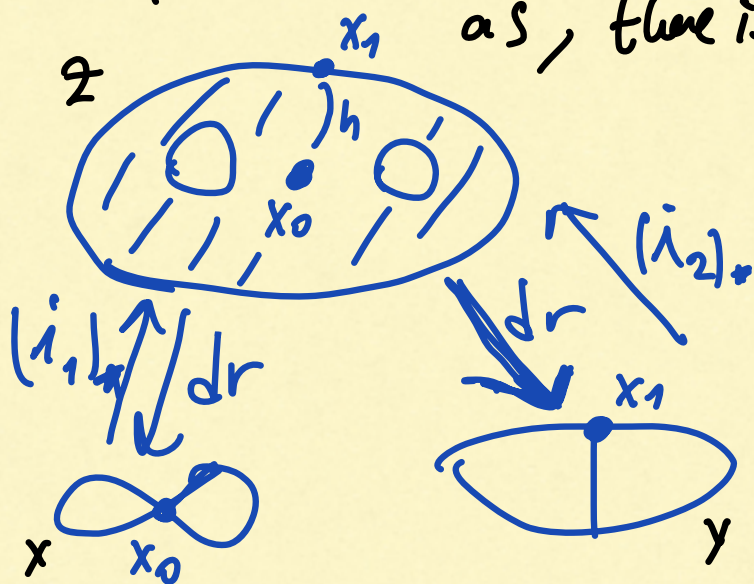
X, Y homotopy equivalence

alternative, tricky approach: As $X \simeq_h Y$

there is some upper space Z , containing both s.t.
(proof of that relies on mapping cylinder)



If X and Y is contained in the same path component of Z then immediate as, there is some path connecting x_0, x_1



and corresponding isom is

$$(i_2)_* \beta_h (i_1)_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, x_1)$$

usual way: $X \simeq_h Y$, telling that

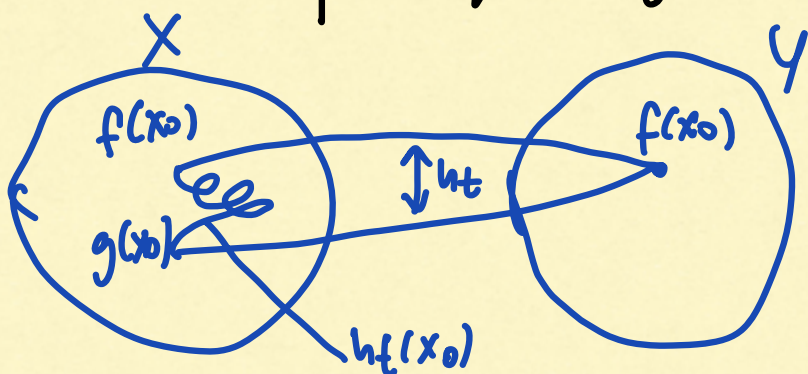
$(f_g) \xleftrightarrow{h_6} Id_X$. What is that info

telling? So far we showed loops can be

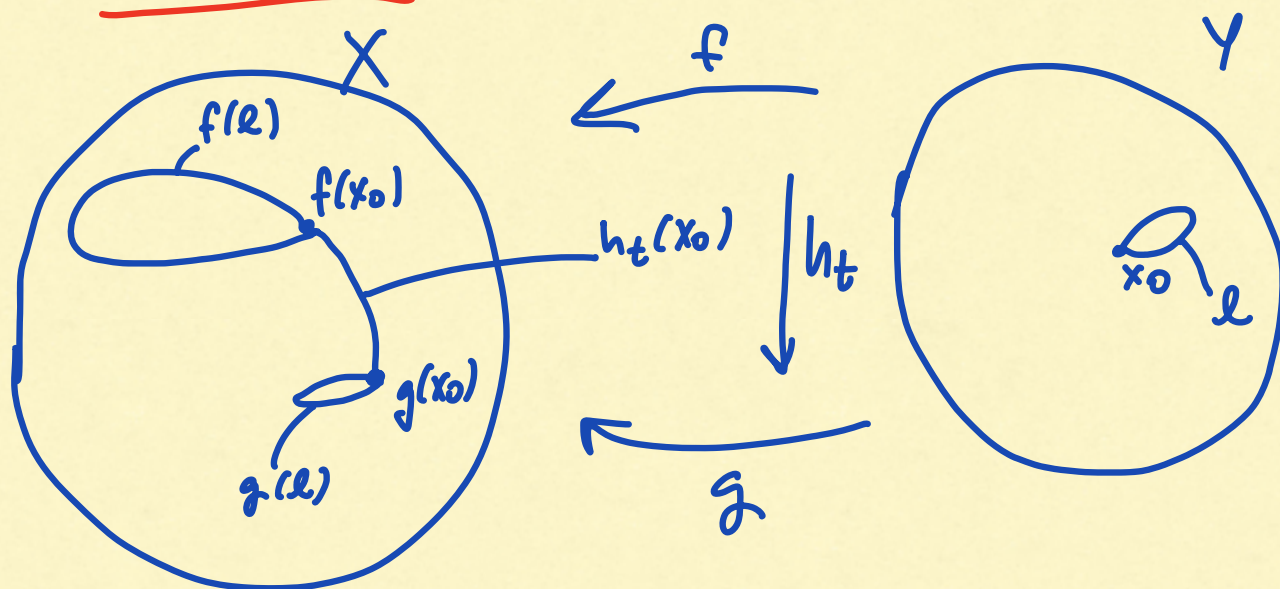
moved with $r: X \rightarrow A \xleftrightarrow{r_6} Id_X$.

Now, lemma 1.19 is, if two maps are homotopic, the induced homomorphisms of them are related by some path.

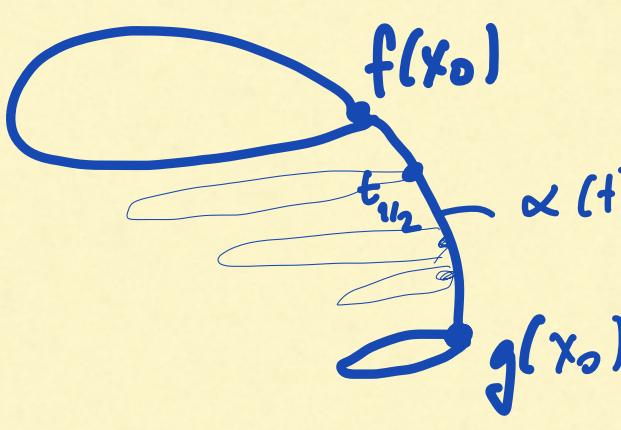
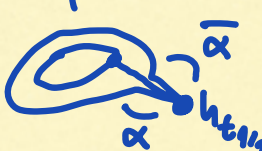
i.e. if $f \stackrel{\sim}{\sim}_h g$, then for some fixed x_0 , $f(x_0), g(x_0)$ can be connected by some path, precisely $h_t(x_0)$!



actually, $f_* = \beta_h g_*$ i.e.



Q: OK, h connects the basepoints, $f(x_0)$ to $g(x_0)$. BUT, is it true that loops that are at these base points path-homotopic? We need to connect them.

$\alpha_t = h_t(x_0)$ is the path with $\alpha: [0, t] \rightarrow X$!

 To have , usual path $\alpha: [0, 1] \rightarrow X$
 reparametrize $\alpha(t) := \alpha(st)$
 for $s \in [0, 1]$. (not that important)
 we will abuse the path, just say $h_t(x_0)$.

Our homotopy connecting loops is,

$$d_t \cdot (h_t(l)) \cdot \overline{\alpha_t}.$$

It is homotopy of loops at $f(x_0)$!

$$\underline{t=0}: \alpha_0(h_0(l)) \overline{\alpha_0} = h_0(l) = f(l)$$

$$\underline{t=1}: \alpha_1(h_1(l)) \overline{\alpha_1} = d_1(g(l)) \overline{\alpha_1} = \beta_{\alpha_1}(g(l))$$

$$\Rightarrow f_*([l]) = \beta_{\alpha}(g_*([l]))$$

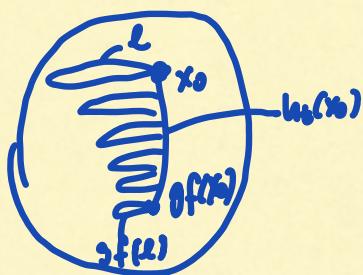
Now, X, Y homotopy equivalent then

$$(gf) \simeq_h Id_X \Rightarrow (gf)_* = \beta_{h_t(x_0)}(Id_X)_*$$

$$= \beta_{h_t(x_0)} Id_{\pi_1}$$

$$= \beta_{h_t(x_0)} \cdot \underline{\underline{[isomorphism!]}}$$

i.e



which says, $(gf)_*$ isom, and f_* injective.
 doing same for $(fg)_*$, f_* surjective,
 with $f_*: \pi_1(X, x_0) \longrightarrow \pi_1(Y, f(x_0))!$
 $[c] \longmapsto [f(c)]$ ■

Interesting example:

Take loops generated by the homotopy
 connecting Id_X to Id_X and fix to
 some base point in order to have a
 closed path, loop i.e

$Id_X \xleftarrow{h_t} Id_X$, $h_t(x_0)$ is loop

as $Id(x_0) = h_1(x_0) = h_0(x_0) = Id(x_0)$.

we have, from previous lemma,

$$Id_X \simeq_a Id_X \Rightarrow (Id_X)_*([c]) = \beta_{h_t(x_0)}(Id_X)_*([c])$$

$$\Leftrightarrow [c] = \beta_{h_t(x_0)}[c]$$

$$\Leftrightarrow [c] = h_t(x_0)[c]h_t(x_0)$$

i.e such loops are in the center!
 $[h_t(x_0)] \in Z(\pi_1(X, x_0))$

